

CLEVELAND-CLIFFS STEEL
Safety Handbook

Effective Date:	<u>11-14-2022</u>	Supersedes:	<u>All Previous Versions</u>
		Standard No.:	<u>CCS_HSS_MAN_001.1</u>
		Follow-Up Review Date:	<u>11-14-2023</u>

APPROVED

K. Dirk Berard Senior Director Health Safety Security	<u>11/14/2022</u>
APPROVER	DATE

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1 Purpose/Scope

- 1.1 This Handbook establishes minimum standards for Personal protective equipment, Ariel lifts and Scissor lifts, Entering Buildings / Departments, Excavations, Floors and Pit Openings Fire Prevention and Firefighting, General Requirements, Housekeeping, Incident Investigation Industrial Vehicles: Operation and Care, Ladders, Scaffolding and Staging, Personal Protective Equipment, Personal Vehicles, Powered Hand Tools and Fixed Equipment, Safe Storing, Loading and Manual Handling of Material, Three-Part Communication, Use and Care of Hand Tools, Use of Tape for Hazard Demarcation, Walkways, Work Areas, Roadways, Stairs and Doors Welding Burning and Cutting, Working at Overhead Locations.
- 1.2 This policy applies to employees of Cleveland-Cliffs Steel business units, contractors, subcontractors and hired labor employees performing work for or on behalf of Cleveland-Cliffs Steel on premises owned, rented or otherwise occupied by Cleveland-Cliffs Steel.

2 Responsibilities and Authorities

- 2.1 This standard was jointly developed with participation from the represented workforce and management.



- 2.2 Each facilities management is responsible for the implementation of this standard.
- 2.3 The Senior Director of Health, Safety and Security is responsible for authorizing this standard.

3 References

3.1 References

- CCS_HSS_STD_001.1 Energy Isolation Safety Standard
- CCS_HSS_STD_002.1 Overhead Crane Safety Standard
- CCS_HSS_STD_003.1 Rail Safety Standard
- CCS_HSS_STD_004.1 Confined Space Safety Standard
- CCS_HSS_STD_005.1 Fall Protection Safety Standard
- CCS_HSS_STD_006.1 Aerial Lifts Safety Standard
- CCS_HSS_STD_007.1 Mobile Equipment Safety Standard
- CCS_HSS_STD_008.1 Tool Safety Standard
- CCS_HSS_STD_009.1 Cutting, Burning, and Welding Safety Standard
- 1910.67 Aerial Lifts and Mobile Work Platforms
- 1910.252 Welding Burning and Brazing
- 1910.176 Material Handling

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- 1910.141 General Environmental Controls
- 1910.132 Personal Protective Equipment
- ANSI Z-89.1
- ASTM-F2413
- ANSI Z.87
- ANSI 105
- NIOSH 42 CFR 84

4 Resources

4.1 None

5 Requirements

5.1 General Requirements

5.1.1 Employees must report to work “Fit for Duty”

- A state which enables the employee to perform assigned tasks competently and in a manner which does not compromise or threatens the safety of health of yourselves or others.

5.1.2 Keep yourself or others outside the “Line of Fire”

- Pinch points, release of hazardous energy, beneath the path of dropped objects, path of mobile equipment etc.

5.1.3 Horseplay, startling, teasing, throwing of objects, scuffling or distracting another employee is strictly prohibited.

5.1.4 Never lean against, sit, or stand, on temporary hand railings or guard railings of any kind.

5.1.5 Before sitting in a chair assure that it is in good repair (including all of the legs) and that is placed on a flat level surface, with all of the legs making contact with the floor surface.

- While sitting in a chair, never tip the chair to cause one or more of the chair legs to lose contact with the floor surface.
- While sitting in a chair, never prop your feet on a surface above the level of the seat or in any other way that will cause the chair to become unstable.
- Chairs with a single post and a base shall have a minimum of 5 legs, with or without casters, originating from the center post

5.1.6 Never dispose of empty aerosol cans or gas cylinders in dumpsters or containers in which they could end up in molten metal, scrap, or be exposed to heat.

5.1.7 Intercoms, PA Systems and Two-way radios are to be used for business purposes only. Non-essential communication and inappropriate language are prohibited.

5.1.8 Cell phones and laptop computers provided by the company and used to conduct business may be carried on the plant property and into operating areas.

- Cell phones with photographic or video capabilities shall not be used for taking photographs or videos on company property unless authorized by Cleveland-Cliffs Steel management.
- Cell phones or electronic devices shall not be used in operating areas. Employees must move to a safe location before using these devices.

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5.1.9 When operating a company-owned vehicle, rental vehicle, leased vehicle or personal vehicle, while performing company business, drivers shall not operate a cell phone, electronic communication device or any other electronic equipment, that is NOT "Hands-Free" equipped.

- The term "Hands-free" equipped is defined as an attachment, add-on, built-in feature or addition to a mobile cellular telephone that, when used, allows the driver to maintain both hands on the steering wheel when receiving, talking and terminating a cellular call. The hands-free accessory can be either permanently installed in the vehicle or may be portable.
- This rule shall apply to company-owned/issued devices or ones owned by the employee.
- If an employee must utilize a cell phone or tablet that are not "Hands-free" they shall drive to a safe location and stop the vehicle prior to activating the device, remaining there until the communication is completed.

5.2 Personal Vehicles

5.2.1 Anyone on a motorcycle shall be wearing an approved motorcycle helmet according to local and state laws and/or plant specific policy.

- Motorcycles are only permitted to be ridden and parked in Cleveland-Cliffs Steel approved locations.

5.2.2 Obey posted signs (speed limit, parking areas, directional signs, etc.)

5.2.3 No personal trailers, boats, campers, etc. allowed on property without written approval by plant general manager or designee.

5.2.4 If a personal vehicle becomes disabled on Cleveland-Cliffs Steel property, employees are not permitted to work on the vehicle, but must call an (outside) Road Service to respond.

- Note: This restriction does not apply to simple repairs, such as changing a flat tire, jump starting a dead battery, and third-party windshield repair.

5.3 Entering Buildings / Departments

5.3.1 Do not go into any department, other than your own, unless your work requires you to do so.

5.3.2 Sign in and out as required by the department prior to entering and after exiting the operations area.

5.3.3 When going into other departments contact the appropriate department person at the sign in location and comply with that department's safety rules, including the wearing of all appropriate personal protective equipment.

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- 5.3.4 Always enter through designated man doors.
- It is only permitted to enter through a roll-up door, if a man door is not present and the door has been pinned, posted and operational control is locked out.
- 5.3.5 When entering a building, stop and assess operational conditions and possible movement of overhead cranes or other mobile equipment before proceeding. Do not assume that crane or mobile equipment operators can see you.
- 5.3.6 Warning signs and warning signal systems have been established in each department. Know and obey the signs and signals of the department where you work and those of any department that your work requires you to enter.
- 5.3.7 Signals - Make sure you know the Emergency Signals and Evacuation Plans in each department where you work.

5.4 Incident Investigation

- 5.4.1 Report all work-related injuries, occupational illnesses, near misses and equipment, product and property damage incidents to Cleveland-Cliffs Steel Management immediately.
- If a work-related injury or illness is not immediately apparent, it shall be reported as soon as the injury or illness is discovered.
- 5.4.2 Employees have the right to report work-related injuries and illnesses; and employers are prohibited from discharging or in any manner discriminating against employees for reporting work-related injuries or illnesses.
- 5.4.3 All involved employees are required to cooperate with management, union safety advocates /coordinators, and other people investigating any incidents or exposures. When asked about an incident with injury/incident/exposure, provide detailed factual information of what was witnessed. This will include:
- All perishable evidence must be preserved
 - Interviews
 - Statements
 - Photos/videos
- 5.4.4 Emergency Reporting Procedures and Emergency Action Plan must be followed.

5.5 Three-Part Communication

- 5.5.1 Three-part communication shall be included in all applicable Job Safety Procedures and when the actions of another person is affected. Applicable Job Safety Procedures shall include, LOTOs, temporary work instruction, crane boarding/ de-boarding requirements etc. or where more than one person or crew is working together on a task.
- 5.5.2 Anytime more than one person or crew is involved in a task that requires understanding and coordination between the two, clear and concise “Initiate-Acknowledge-Confirm (IAC)” communication shall be utilized in the following manner:
- Employee 1 – Initiates communication.
 - Employee 2 – Acknowledges by repeating the message.
 - Employee 1 – Confirms that Employee 2 received the correct message.
- 5.5.3 All work shall PAUSE if any of the three parts of IAC are not clearly understood (i.e., employee 1 does not initiate the communication, employee 2 does not acknowledge the initial communication, or employee 1 does not confirm correct understanding from employee 2).
- 5.5.4 Avoid using single word responses that could be easily confused such as “Go” and “No” or using phrases that end in an action that rely a single word such as “Clear” or “Not clear”. Use words like negative, positive, or phrases that repeat the emphasis sentence such as “We are not, I repeat, not clear.”
- 5.5.5 Any confusion in communications should result in a pause of work to clarify what is being said and what actions need to occur.

5.6 Housekeeping

- 5.6.1 Keep your work area clean and clear. Pick up and dispose of debris as it accumulates. Return hand tools and equipment to their proper storage places when finished with them.
- Keep all walkways, aisles, stairs, and platforms clear of obstructions. Do not pile materials in any area where people must walk. Be sure that materials which are stacked or piled near walkways do not project into aisles, stairs or platforms and that they are properly stacked so they will not topple or fall.
 - Do not obstruct access to emergency exits, fixed ladders, stairways, electrical switches, shutoff valves, firefighting or rescue equipment and safety eyewash fountains or safety showers. Do not store any material in, on, or in front of switch boxes or other electrical equipment.

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- 5.6.2 When you have completed repair, construction, or maintenance work of any kind, take time to clean the area. Pick up and remove all waste and salvaged materials. Return all hand tools and equipment to proper storage. Remove all loose objects after completing overhead work. Replace all covers, guards and/or access panels.
- 5.6.3 Containers shall be provided for the collection and separation of waste, trash, and other refuse. Containers used for oily, flammable, or hazardous wastes (such as caustics, acids, harmful dust, etc.) shall be equipped with covers and labeled. Garbage and other waste shall be disposed of at frequent and regular intervals. Never dispose of trash or non-metallic waste in scrap metal containers.
- When working with, moving or cleaning materials such as metal, wood, glass, etc., do not use your hands, forearms, or other parts of your body to remove slag, slivers, chips, dust, etc. from materials or surrounding areas.
- 5.6.4 Each plant shall develop a housekeeping program, which at a minimum must include a monthly area housekeeping inspection.

5.7 Personal Clothing / Jewelry

- 5.7.1 Employee's bodies shall be covered during work hours to comply with department and plant regulations. Only approved apparel, worn in the proper manner shall be permitted in the plant.
- 5.7.2 Pants must extend at least to the ankles.
- 5.7.3 Long sleeved shirts are required in all operating areas with the sleeves down and buttoned at the wrist. The shirt shall be tucked into the trousers and fully buttoned up the front to the first button below the collar button. The collar button shall be buttoned when directed by specific Department Procedures. Exceptions to this rule:
- This rule does not apply to non-operating areas such as office areas, break rooms, general office buildings, etc.
 - Permitting rolled up long sleeves in operating areas (Machine Shops. Etc.), shall be documented in writing and approved by the Plant Manager and Safety Department.
- 5.7.4 Work clothes shall not be cleaned in flammable liquids, such as gasoline, kerosene, or similar fluids.
- 5.7.5 Synthetic clothing including under clothing, that is not made of approved fire retardant or fire-resistant materials, shall not be worn in hot metal areas or when exposed to burning, welding, flame, high radiant heat areas or potential arc flash.
- 5.7.6 Do not wear clothing that is loose fitting, torn, or ragged, too long (pant legs/shirt sleeves) or with attached hoods (i.e., hooded sweatshirts) to avoid being caught on projections or in moving machinery.

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- 5.7.7 Permanently affixed hooded garments are not permitted unless a documented hazard assessment has been conducted and requirement is specified in a Job Safety Procedure. (I.e., hooded tap coats, disposable coveralls, etc.)
- 5.7.8 Wearing of finger rings (including silicone or rubber rings), loose-fitting wrist bracelets, exposed dangling necklaces, and dangling earrings or nose rings, other than stud type earrings, or having gauged ear lobes that are open, are prohibited in all operating areas.
- Stud type earrings are permitted in the operating areas as long as they are single studs and fit directly over the opening in the ear lobe and do not project up or down in any manner.
 - Gauged ear lobes that have the hole plugged with a solid blank or covered with tape or a Band-Aid are permitted.
 - All earrings and gauged lobes must be covered with tape or a Band-Aid while performing any type of electrical work.
- 5.7.9 Employees with shoulder length or longer hair, or long facial hair (beards) shall be maintained / secured so that it does not create a hazard.

5.8 Personal Protective Equipment

- 5.8.1 Cleveland-Cliffs Steel employees shall be trained on the proper inspection and use of all Personal Protective Equipment (PPE) issued to them.
- Unauthorized alterations of personal protective equipment in any way are strictly prohibited.
- 5.8.2 All Personal Protective Equipment (PPE) must follow manufacturer's recommendations and requirements.
- 5.8.3 Always inspect the PPE before use and replace if needed. Report defects to Cleveland-Cliffs Steel Management immediately.
- 5.8.4 Cleveland-Cliffs Steel employees are not permitted to use PPE on Cleveland-Cliffs Steel property that were not approved and provided by Cleveland-Cliffs.
- 5.8.5 Job Safety Procedures or Department Rules shall specify the PPE required for each job. If the PPE is not identified, contact Cleveland-Cliffs Steel Management to confirm PPE requirements before starting the task or if the task's conditions have changed and may require additional PPE.
- 5.8.6 Persons working within 10 feet of a body of water, 4 feet or greater in depth, shall wear an approved PFD (personal floatation device), if not protected by handrail or other fall protection.

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5.8.7 Protective Helmets

5.8.8 Only Cleveland-Cliffs Steel approved Protective helmets conforming to ANSI Z-89.1 Industrial Standard shall be worn. Your protective helmet shall:

- Be worn with bill to the front.

Unless an approved exception is documented in a hazard assessment and Job Safety Procedure.

- Without altering or wearing anything that would affect the suspension system.
- Hats (baseball type hats, tassel caps, toboggans, etc.) that may affect the fit of the helmet shall not be worn under the protective helmet. Welders' caps, liners or other tight fitting garment that do not affect the hat hard suspension are permitted to be worn under the helmets.
- Without painting, drilling, or otherwise altering the helmet shell.
- With your first and last name printed or labeled onto the front of the helmet.
- With reflective safety tape as follows: · At least a three (3) inch long strip on each side and back of the helmet placed approximately one (1) inch up from the brim of the hat.
- Hardhat or suspension cracked, chipped, burnt, or otherwise suspected to be damaged shall be taken out of service immediately.
- Employees operating or riding in enclosed crane cabs or mobile equipment are not required to wear hard hats unless directed to do so by a specific Job Safety Procedure.
- A red flashing light placed where visible is required to enter designated coil fields, low visibility situations and/or as specified in a Job Safety Procedure. (I.e. roll storage racks, slab yards, batch annealing operations, etc.)

5.8.9 Metatarsal Boots

- Only Cleveland-Cliffs Steel approved six inch or higher, safety toe, safety boots with permanently attached or internal metatarsal guards conforming to ANSI Z-41 Industrial Standard in good repair, properly laced and tied shall be worn by Cleveland-Cliffs Steel employees.
- Cleveland-Cliffs Steel employees are only permitted to obtain metatarsal boots through Cleveland-Cliffs Steel approved metatarsal boot vendors.
- Rubber type metatarsal boots are not permitted unless required by the Job Safety Procedure.
- Temporary metatarsal guards are not permitted.

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5.8.10 Safety Glasses

5.8.11 Only Cleveland-Cliffs Steel approved safety plastic framed glasses with permanently attached side shields or wrap around design which meet the ANSI Z.87 Industrial Standard shall be worn by Cleveland-Cliffs Steel employees.

- Employees are only permitted to obtain prescription safety glasses through Cleveland-Cliffs Steel approved eye glass vendors.
- Additional eye protection, such as face shields, welding helmets, goggles, etc., may be required for specific jobs. Always wear the type of eye protection prescribed for the job as specified by the Job Safety Procedure.
- Employees operating or riding in fully enclosed crane cabs or mobile equipment with all windows closed are not required to wear approved safety glasses unless directed to do so by a specific Job Safety Procedure.
- Sunglasses, mirrored sunglasses, transition glasses, etc. are not permitted to be worn inside the plant.

Prescription exemptions and or medical conditions must be documented in writing and be approved by the employee's doctor / plant medical and the Safety Department

Employees performing tasks identified in the Job Safety Procedure requiring dark lenses shall be provided the appropriate glasses, clip-ons, flip downs, goggles, face shields, or hoods.

5.8.12 Face Protection

5.8.13 Only Cleveland-Cliffs Steel approved face shields which meet the ANSI Z.87 Industrial Standard shall be worn.

5.8.14 The material constructing the face shield shall be selected based on the potential hazard. Solid material (polycarbonate, acetate, etc.) face shields shall be worn when the hazards is either liquid, molten, arc flash, impact or steam.

5.8.15 Face shields shall be worn in addition to approved safety glasses or goggles.

5.8.16 A face shield shall be worn in the down position if the work activity includes one or more of the following hazards:

- Handling metal or plastic banding while banding coils or other materials, picking up and carrying loose bands or when cutting bands from coils, bundles, pallets, etc. (> 6 inches in length).
- Employees within an arms' length of metal ends, samples, scrap or other materials that can whip or spring, striking the face (i.e. edge trim, weld scarf, scrap bales, flashing).
- Handling full width samples or scrap.

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- Exposure to flame, sparks, ultraviolet radiation (welding) arc flash or radiant heat as defined in a hazard assessment.
- Exposure to molten metal (zinc, aluminum, steel, iron, etc.).
- Exposure to acidic, caustic, hot and/or cryogenic fluids
- Spraying with pressurized (>25 psi) liquids, air and mixtures. (I.e. refractory gunning, pressure washing, cleaning via water hose, abrasive blasting, etc.).
- Exposure to steam
- When performing work that could cause deposited dust or metal fines to drop down onto the employee(s).
- Grinding, chipping, striking or cutting operations that cause flying sparks, chips or particles.
- While troubleshooting and/or performing maintenance on hydraulic, pneumatic, steam, or any other line that is under pressure.
- As required by the Job Safety Procedure or department rules.

5.8.17 When working in chemical areas where the procedure requires a chemical barrier suit, the face shield shall be equipped with a chin cup protector to enhance protection to the face and neck. In addition, goggles shall be worn under the face shield for added protection to the eyes.

5.8.18 Respiratory Protection

5.8.19 Only Cleveland-Cliffs Steel designated NIOSH approved respiratory protection shall be worn.

- If wearing a respirator is required, you must be medically evaluated, fit tested and properly trained.
- Facial hair or stubble shall be trimmed or shaved to ensure an adequate seal and proper function of the respirator
- Respirators must be stored per the manufacturers requirements
- Employees shall inspect their respirators prior to use.
- Employees shall conduct field verification (positive and negative) fit checks prior to entering the respiratory required work area.

5.8.20 Voluntary use of respiratory protection requires the employee to be provided a copy of Appendix D to Sec. 1910.134 (Mandatory) Information for Employees Using Respirators When Not Required under the Standard.

5.8.21 Hand Protection

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- Before handling any object, substance, or tool, approved hand protection shall be selected and worn to provide the required level of protection.
- Hand protection shall be selected based on an evaluation of the performance characteristics of the equipment relative to the job tasks to be performed and the type of hazard/hazards associated with the tasks.
- Approved and designated cut resistant gloves shall be properly worn when the employee is within an arm's length of cut hazard from coils, metal strips, banding, scrap, etc. or as directed by the Job Safety Procedure or department rules.
- At a minimum, approved leather palmed puncture resistant gloves shall be properly worn when handling wire rope, wire slings or other materials with a potential puncture hazard or as directed by the Job Safety Procedure or Department Rules.
- Hand protection shall be designated in a Job Safety Procedure or by department rules and shall be worn as required.
- Hand protection shall be worn as designed by the manufacturer and shall not be altered from the original design.

5.8.22 Arm Protection

5.8.23 All employees wear Cut resistant sleeves when within arm's length of coils, samples, strip, scrap, or other cut hazards; or when directed by a Job Safety Procedure.

- Cut resistant sleeves are not required after a PPE hazard assessment has been performed and there is no exposure to cut hazard.
- Cut resistant sleeves shall be worn with the thumb through the thumbhole and finger(s) through the finger hole(s) and worn under gloves when required
- The Cut resistant sleeves must be extended above the elbow, where they shall be secured with the clips or in such a manner that the Cut resistant sleeves will remain in place.

5.8.24 Torso and Leg Protection

- Full length cut-resistant chaps shall be worn whenever the employee is within an arm's length of coils, metal strip, scrap, banding, or other similar cut hazards.
- Full length cut resistant chaps shall be worn when walking/working in designated coil fields or as required by a Job Safety Procedure.

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- Cut-resistant aprons shall be worn when there is a potential to strike the torso against protruding metal strip, scrap, banding, or other similar cut hazards or as required by a Job Safety Procedure.
- Aprons shall not be worn when there is a potential of being caught in machinery.

5.8.25 High Heat Exposure Clothing

- Approved aluminized coat, or equivalent material, completely secured, shall be worn to protect the base of the neck, torso, arms and upper legs from radiant heat, molten metal, and metal red zones or as required by a Job Safety Procedure and/or department rules and shall extend at least 2 inches over the spats.
- Approved aluminized, or other equivalent material, spats shall be worn to protect from the top of the foot to the knee from radiant heat, molten metal and sparks or as required by a Job Safety Procedure.
- The spats may be worn under the outer garment or over the outer garment in the following manner:

Non-Aluminized spats worn under approved flame-retardant pants or coveralls shall be worn over approved under garments to provide additional protection from the hot work hazards and to prevent skin irritation.

Aluminized spats worn over approved issued uniform shall have a minimum of one inch of the pant leg blousing out and over the top of the main body section of the leggings

- Approved aluminized or equivalent material jacket with reflective tape with hood, completely secured and hood over helmet, shall be worn to protect the head, neck and torso of the railroad crew within 30 feet of torpedo cars transporting molten iron.
- The aluminized jacket requirements do not apply to burning or brazing with a hand torch or hand held welding or air arcing unless specifically required by a Job Safety Procedure.
- Approved aluminized neck shroud or equivalent material shall be properly worn in conjunction with an approved face shield and carbonex type hood when there is a molten metal splash hazard or as required by the Job Safety Procedure.

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- Approved aluminized or equivalent material beehive type hood shall be worn when the combination of a face shield, aluminized shroud and carbonex type hood does not provide adequate protection from exposures to molten metal, sparks, radiant heat, flame and/or steam or as required by the Job Safety Procedure.
- Approved long cuff gloves with cinches when worn over the aluminized or equivalent material coat shall be cinched tight.
- Approved or equivalent material gloves without cinches shall be worn under the aluminized or equivalent material coat sleeves and the coat sleeves with Velcro closure or snaps shall be tightened or snapped over the gloves.

5.8.26 Minimum PPE

- Cleveland-Cliffs Steel issued uniform:
- FR ankle length pants, FR wrist length shirt (Hot metal uniforms pant Hot metal uniforms shirts in all department's where molten metal is present)
- Hard hat
- Safety eyewear
- Metatarsal boots with safety toe
- Appropriate gloves for the task
- Hearing protection where required

5.8.27 Coil fields or working within arm length of a coil

In addition to minimum PPE, when interacting with coils where there is a potential to strike against or be struck by protruding metal strip, scrap, banding, or other similar cut hazards or as required by a Job Safety Procedure.

- Cut resistant chaps
- Cut resistant bib/apron
- Cut resistant sleeves
- ANSI cut level 5 gloves
- Face shield

5.8.28 Hot metal PPE

In addition to minimum PPE, when interacting with molten metal or positioned in a designated molten metal red zone area, the following PPE is required:

- Base layer (Carbon armor, CarbonX, DRIFIRE)* Optional

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- Aluminized or equivalent material Tap coat
- Aluminized or equivalent material Spats
- Aluminized or equivalent material Gloves
- CarbonX hood
- Aluminized or equivalent material neck drape
- Face shield
- Aluminized or equivalent material Beehive hood if job task requires based on the hazard assessment or Job Safety Procedure as required.

5.8.29 Hot Work

In addition to minimum PPE, when conducting Hot Work activities the following PPE is required:

- Double layer FR (FR welding jacket or leather welding jacket)
- Face shield / Welding helmet (attached to hard hat or integrated into the design)
- FR hood (if hot work is being performed above shoulder height)
- Hot work or welding gloves
- Leather or Equivalent spats

5.8.30 Any employee within 6' shall wear the exact PPE as the employee performing the welding, burning, and cutting.

5.9 **Fire Prevention and Firefighting**

5.9.1 Include emergency response to fires

- In addition to the information below, sites are required to develop and maintain emergency action plans that include emergency response to fires.

5.9.2 Prevent Fires

5.9.3 Follow fire prevention rules for your area and be alert to potential fire hazards throughout the plant. Be aware that the danger of fires is higher in certain areas of the plant and special precautions must be taken. Know the proper way of responding to a fire and how to turn in an alarm or initiate the sites emergency notification procedure in the area where you are working. Know the emergency escape routes and the designated assembly points should evacuation become necessary.

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- Always obey local smoking regulations.
- Do not allow rubbish or flammable materials to accumulate. Oily waste, rags, gloves, etc., shall always be placed in receptacles provided for them.
- Proper storage of flammable liquids, aerosol cans, etc.
- Change your clothes immediately if they become soaked with oil, gasoline, kerosene, naphtha, or other flammable or combustible materials. Stay far away from portable heaters, welding sparks, hot metal, and other sources of fire if your clothing has been exposed to flammable or combustible liquids.
- Do not light oil or gas furnaces, stoves, portable heater, or other heating equipment unless you have been properly instructed and authorized. Never deviate from specific written procedures when lighting furnaces, stoves, heaters. Etc.
- Flammable liquids (flash point under 100°F) may be used only where there are no open flames or other sources of ignition within any possible path of a vapor trail. Do not dispense a flammable liquid into another container unless the nozzle and container are properly grounded and labeled. Flammable liquids shall be disposed from approved safety cans only.

5.9.4 Firefighting Equipment

- Fire protection system shall not be disabled without written authorization and notification to affected employees in the area.
- Only operate firefighting equipment that you have been trained on.
- Ensure firefighting equipment is properly inspected and maintained
- Firefighting equipment must not be used for any purpose except to put out fires. Do not obstruct aisles or block the paths to fire plugs, extinguishers, hose stands, and other firefighting equipment.
- Fire extinguisher locations must be clearly marked and identified.
- Familiarize yourself with the location of the nearest fire extinguisher or fire hose in the area where you work.

5.10 **Welding Burning and Cutting**

5.10.1 For Additional information and full scope of requirements reference CCS_HSS_STD_009.1 Cutting, Burning, and Welding Safety Standard

5.10.2 Hot Work Permitting

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- Specific areas in the plant will be designated for hot work such as burning, welding, and grinding. If an area is not designed and designated for hot work; a hot work permit must be filled out before hot work can begin. The permit may require the use of fire blankets, other means of spark containment, removal of fire hazards, air quality tests, fire watch, etc., to protect other employees and surrounding property.
- Supervision will designate an individual responsible for authorizing burning, welding, grinding, and other hot work operations in areas not specifically designed or approved for such operations. The work area must be inspected, all portions of the Hot Work Permit completed, and the hot work permit signed by the authorized employee.

5.10.3 Work must be shielded when welding or burning to protect others from the arc flash, or ultraviolet rays.

- If it is necessary for you to be in an area where welding or cutting is being done, exercise caution. Never look directly at the intense light generated by welding or cutting. Check your clothing to be sure you are not carrying sparks which might set your clothing afire. Grease and/or oil on your clothes are especially dangerous if you are working around oxygen. Oxygen can cause the grease or oil to ignite.

5.10.4 Always check burning equipment for leaks before use with a visual inspection and leak detection.

5.10.5 Gas and oxygen hoses need to have flashback prevention and anti-kink strain relief devices installed.

5.10.6 Cylinders Containing Compressed Gases

- Oxygen, fuel gas, and other compressed gases must be considered dangerous fire hazards because they support or increase the rate of combustion. When handling cylinders containing any kind of compressed gas, observe the following rules and specific instructions of your supervisor: If clothes are contaminated with grease, or oil, or solvents, they must be changed before performing any burning or welding.

5.10.7 When burning, the torch shall be extinguished when it becomes necessary to reposition yourself, the compressed gas cylinders or the hoses, around or over obstructions.

5.10.8 When burning, the torch shall be extinguished when it becomes necessary to remove the twist from the oxygen and fuel hoses.

5.10.9 Disposal of Propane and Calibration Gas Bottles.

- Depressurize each bottle to atmospheric pressure and clearly mark the bottle EMPTY with a Semi Permanent Marker. The bottles can then be placed into a trash disposal container or returned to the vendor for re-use.

5.10.10 Transporting

- Caps shall be in place on compressed gas cylinders unless they are in use. Always replace the caps on full or empty cylinders after regulators have been removed. It is especially important to have the caps in place while transporting cylinders to protect the valve.
- Compressed gas cylinders must be transported in special containers or carriers provided for that purpose. At a minimum the carriers shall have a retainer system to secure the cylinder from falling during transport. Torch carts shall have a divider between the oxygen and fuel gasses.
- Do not handle cylinders with crane slings or magnets. Use good judgment when handling or moving cylinders and do not lift beyond your capabilities.
- Compressed gas cylinders must always be securely tied off in a vertical position.
- Do not transport compressed gas cylinders in passenger elevators.

5.10.11 Storage

- Compressed gas cylinders shall be stored with regulators removed and the cylinder caps installed and tightened.
- Compressed gas cylinders shall not be stored near stoves, furnaces, near molten metal, portable heaters or in any area where they will be subjected to high temperatures. They must not be stored where they can make unexpected contact with electrical equipment or conductors if they should slip, roll or fall. Do not store cylinders in entrance/exit areas, or in stair wells.
- Oxygen cylinders must not be stored in a confined or enclosed space with fuel gas, liquid flammables, oils, or grease. A leaking cylinder can cause spontaneous combustion of these materials.
- Acetylene cylinders must always be stored vertically and used vertically with the valve end up.
- Oxygen shall be separated from fuel-gas cylinders or combustible materials at a minimum of 20 feet or by a noncombustible barrier at least five feet high having a one-half hour fire resistance rating.

5.10.12 Using Cylinders

- Never use oil or grease to lubricate any part of an oxygen, acetylene or other gas cylinder or regulator. Do not touch them if you have oil or grease on your hands or gloves.
- If a leaking cylinder is found, remove it to the outside immediately and keep it away from any flames or combustible materials. Notify your supervisor.

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- Hoses for fuel gas-oxygen rigs - GREEN hose is for oxygen; RED hose is for all fuel gas. Never use compressed air to blow out these hoses. To prevent tripping hazards and hose damage, keep excess hose lengths coiled and lines arranged in orderly fashion.
- Never bring cylinders into, or store cylinders in confined spaces, unventilated rooms, or other closed quarters. Storage areas must be open or ventilated to prevent a build-up of gas.
- Hoses which are damaged shall be cut in two and taken out of service to prevent injury. Branching or using T-fittings in oxygen or fuel gas hoses is prohibited.
- Each user shall visually inspect hoses for wear and loose connections. If defects are found the hose shall be removed from service and not used.
- Prior to each use, the working end connections and fittings of the hose shall be tested for leaks by immersing or spraying the pressurized hose with an approved leak detection solution. If bubbles, or leaking is detected the hose shall be removed from service and not used.
- Wire brushes, scrapers, or other appropriate hand tools shall be used to clean a surface before or after burning, cutting, or making weld repairs.
- Oxygen, or welding shield gases (Such as Argon, CO2, etc.) shall not be used to clean surfaces before burning, cutting or making weld repairs.

5.11 Powered Hand Tools and Fixed Equipment

5.11.1 For Additional information and full scope of requirements reference CCS_HSS_STD_008.1 Tool Safety Standard

5.11.2 All powered tools and equipment must be inspected before each use.

- All tools must be physically and mechanically sound, and appropriate for the job.
- If any defects are detected, the tool must be removed from service immediately.
- Only properly trained and authorized personnel are permitted to operate these types of equipment.

5.11.3 Guards, shields or safety devices specifically designed for portable power tools must be installed, properly functioning, and in good condition before use.

5.11.4 Electric tools or equipment with broken three-prong grounding plugs must be taken out of service.

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- All portable cord and plug electric tools must be connected to G.F.C.I. (Ground Fault Circuit Interrupter).
- Extension cords must be inspected for kinks, worn insulation, and exposed strands of wire before use.
- Extension cords must be protected from physical damage and installed to be out of walkways, stairways, aisles, and areas where they present a trip hazard.

5.11.5 Pneumatic Power Tools and Hose.

5.11.6 Pneumatic power tools must be connected only to designated plant air lines.

5.11.7 They shall be secured to the hose by a positive means to prevent the tool from becoming accidentally disconnected.

- By using safety clips, pins or retainers.
- Anti-whip connections shall be used to connect hoses.
- Automotive radiator type hose clamps are not permitted on air hoses.
- Never direct compressed air toward yourself or anyone else for any reason.
- Never hook up to an unidentified line.

5.11.8 Grinders

- Never use pedestal grinder or portable grinders without the use of face shield and approved eye protection.
- A pedestal grinder tool rest must not be more than 1/8" away from the grinding wheel and the upper tongue guard not more than 1/4" from the grinding wheel.
- Never use pedestal grinder that is not anchored to the mounting surface.
- You should not use or install any grinding wheel unless you have been properly trained in its use and authorized by your supervisor.
- Gloves and loose clothing shall not be worn when using a pedestal or bench grinder.

5.11.9 Hydraulic tools

- Hydraulic powered tools and jacks are required to be visually inspected before and after each use.
- The rated load shall be legibly and permanently marked in a prominent location on the jack.

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- Hydraulic powered tools and jacks shall not be used in a manner that will exceed the rated load or any other capacity of the tool.

5.11.10 Powered hand/fixed tools

- Do not operate equipment that you have not been trained to use.
- Powered hand tools equipped with side handles shall be operated with both hands holding the tool to control kickback or twist.
- Do not use the tool if the handles are defective or missing.
- When using powered hand tools or fixed tools, the work piece shall be secured to prevent movement while cutting, drilling, grinding etc. Hands or parts of the body will not be used to secure the work piece.
- Gloves, and loose clothing shall not be worn when operating powered fixed equipment, such as drill presses, milling machines, lathes, pipe threaders or other rotating equipment, where the point of operation, ingoing nip points or caught-on hazards are accessible (within arm's reach) to the operator.
- Gloves are required before or after machining when handling any sharp and/or hot objects, such as drill bits, milling bits, cutting tools, etc. and when handling the work piece being machined.

5.11.11 Maintenance activities require the powered fixed equipment to be shut-off and LOTO and the employees performing the maintenance shall wear the appropriate gloves

5.12 **Use and Care of Hand Tools**

5.12.1 For Additional information and full scope of requirements reference CCS_HSS_STD_008.1 Tool Safety Standard

5.12.2 All hand tools must be inspected each time before use.

5.12.3 Never use a defective tool or leave defective tools where others might use them.

5.12.4 Never use tools and equipment for purposes for which they were not designed or intended.

5.12.5 Tools shall be transported in toolboxes, tool belts, tool pouches, etc. designed for such use. Do not carry hand tools in pants or shirt pockets unless designed for pocket carry.

5.12.6 Never throw tools, equipment, or parts from one location to another, from one employee to another, or from one level to another level.

5.12.7 Do not leave tools, equipment, or materials in elevated places from which they may fall or be knocked off.

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5.12.8 Do not force tools beyond their capacity or use "cheaters" to increase their capacity.

5.12.9 Do not fabricate hand tools. Do not modify or in any way alter the design of existing hand tools.

5.12.10 Only self-retracting utility knives are permitted to be used for work duties. Unless a specific use of a non-retracting knife is defined in Job Safety Procedure.

- Do not use personal pocketknives as a tool.
- Always cut in the direction away from yourself.
- Wire stripping tools shall be used when preparing electrical wire.
- Alternate tools, other than open blade knives, shall be used to cut materials whenever possible.

5.12.11 Holding tools

- Hands shall not be used to hold or position a tool or object that is being hammered by another employee. Holding tools, jigs, or other positioning devices must be used for this task.
- Hands shall not be used to hold a chisel, punch, or any other tool that is being struck by a hammer. A holding tool, with a handle extension, specifically designed to hold a chisel, punch, etc. shall be used whenever striking or hammering.
- Hand Tools (i.e., channel locks, vice grips, pliers) not specifically designed to hold a chisel, punch or other hammered tools shall not be used for this purpose.

5.13 Industrial Vehicles: Operation and Care

5.13.1 For Additional information and full scope of requirements reference CCS_HSS_STD_007.1 Mobile Equipment Safety Standard

5.13.2 Industrial Vehicles includes all powered industrial trucks (PIT), fork trucks, tractors, platform lift trucks, motorized personnel carts, material handling wheeled or tracked Industrial vehicles and other specialized industrial trucks powered by electric motors or internal combustion engines used exclusively in the industrial setting. The term "Industrial Vehicle" will be used to mean all of the above.

5.13.3 Only employees authorized and trained are permitted to operate or ride in Industrial Vehicles.

5.13.4 All Industrial Vehicles that are 8 feet or wider shall be equipped with a backup camera.

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- Large Industrial Vehicles that are 8 feet or wider which are short term rentals (less than 2 weeks) that are not equipped with a backup camera shall require a spotter.
- 5.13.5 At the beginning of each work shift, operators shall inspect and test their assigned Industrial Vehicle to determine if it is in safe operating condition.
- A daily checklist shall be completed and signed by the operator.
- 5.13.6 If any inadequacies in the operation, nonfunctioning safety devices or other deficiencies effecting safe operation of the Industrial Vehicles are recorded or noted at inspection, the operator will make immediate notification to the appropriate manager or designee.
- Pre-operational check list should have critical and non-critical items identified, if any critical items are identified as deficient, the equipment shall be taken out of serviced and repaired before use.
 - If pre-operational check list does not have identified critical items, then all items are critical.
 - The defective equipment shall not be operated until the repairs are made.
- 5.13.7 Authorized Industrial Vehicle operators shall abide by all traffic regulations and be alert for pedestrians.
- 5.13.8 Do not stay in the cab or truck while loading or unloading unless necessary for the operation of the industrial vehicle.
- 5.13.9 Unattended Industrial Vehicles shall have the engine shut off and parking brake set or block the wheels to prevent unexpected movement or in accordance to written Job Safety Procedure.
- 5.13.10 Industrial Vehicle operators must not permit unauthorized riders on their Industrial Vehicles.
- 5.13.11 Riders are not permitted to ride in the back of pickup trucks or other Industrial Vehicles that do not have specifically approved seats and seat belts.
- 5.13.12 Do not jump off of a moving Industrial Vehicle.
- 5.13.13 Do not ride in an Industrial Vehicle with unsecured loads.
- 5.13.14 When riding in Industrial Vehicles always keep your body parts inside the running lines of the Industrial Vehicle.
- 5.13.15 Pedestrians must be aware of plant Industrial Vehicle movement when walking through buildings or yards.
- 5.13.16 Do not assume the Industrial Vehicle operator can see you. Be prepared for sudden turns or the Industrial Vehicle backing up.

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5.13.17 Areas where industrial vehicles are required to back into buildings, or have limited clearance and visibility must have a hazard assessment conducted to determine the need for controls to eliminate employees from being in the line of fire (guarding, light systems or use of a spotter, etc.)

5.13.18 If a Spotter is used, they shall wear a high-visibility vest for identification and have an air horn or other device to warn unauthorized people away from area.

- The Industrial Vehicle operators shall maintain visual and/or radio communication with the Spotter before initiating any movement of the Industrial Vehicle.
- If visual contact and/or radio communication is lost or restricted, no movement shall occur until contact and/or radio communication is restored.
- The Industrial Vehicle operator shall not move the Industrial Vehicle until given direction to do so by the Spotter and the Spotter is a minimum of 10 feet away from the equipment.
- A Spotter shall assure that all equipment and people are in the clear before giving direction to move the Industrial Vehicle.
- Signs shall be posted at each truck door. The sign shall indicate the door number, the open height of the door, sound horn prior to entering and or spotter is required or not.

5.13.19 Employees working within mobile equipment, shall be trained on the movement capabilities of the equipment to include the swing radius, blind spots, etc. Whenever possible, maintain a minimum distance of 10 feet from the mobile equipment while it is moving.

- If the minimum 10ft distance cannot be maintained, three-part communication must be established between employees working in an area and the operator of the powered industrial vehicles.

5.13.20 Operators of plant Industrial Vehicles are not to transport loads which are unstable or above the rated capacity of their Industrial Vehicles.

5.13.21 Mobile cranes (Excluding: carry decks type crane) are not to be used to transport materials on the hook to and from the job site.

5.13.22 Never use a fork lift or similar Industrial Vehicle as a working platform or allow yourself to be hoisted from one level to another unless a specially designed platform is provided and you have been authorized to use the Industrial Vehicle in this way. The platform must be secured to the fork lift, and the fork lift operator must remain at the controls at all times while the platform is in use.

5.13.23 When a powered industrial truck is left unattended, the load lifting device shall be fully lowered, controls neutralized, power shut off and brakes set. Wheels shall be blocked if the truck is parked on an incline.

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5.13.24 When an Industrial Vehicle is being refueled or serviced, the motor must be shut off.

5.13.25 The Industrial Vehicle being refueled must be attended the entire time during the fueling process.

- All fuel loading nozzles on Cleveland-Cliffs Steel property shall have the trigger hold down device removed.
- Do not smoke or use cell phone while re-fueling.
- No object shall be placed in the handle that will defeat the automatic shut-off at any time during the fueling.

5.13.26 Cleveland-Cliffs Steel employees shall use approved rolling step platforms with handrails to climb onto and off truck and trailer beds.

5.13.27 Operators shall use access ladders and provided steps and/or ladders to get on or off the equipment.

5.13.28 Do not jump from the Industrial Vehicle to the ground.

5.13.29 Slow the Industrial Vehicle down and give adequate warning by sounding the horn or audio warning alarm signal when entering or exiting buildings, or when approaching corners, blind spots, cross aisles, pedestrians, or where employees are working.

5.13.30 Industrial Vehicles shall not be parked closer than ten feet from the outer rail of the nearest railroad track.

5.13.31 Operators shall be aware of overhead crane movements. Maintain a safe distance with your Industrial Vehicle when crane movements are being made.

5.13.32 When following behind large Industrial Vehicles (heavy haul, tractor/trailers, coil trucks, dump trucks, flush trucks, spray trucks, sweeper trucks, front end loaders, etc.) stay back far enough to allow you to see the mirrors of the equipment you are following at all times.

- A minimum following distance of 50 feet shall be maintained behind moving large Industrial Vehicles and at stop signs.

5.13.33 Seat belts must be worn by Industrial Vehicle operators and passengers during operation. If the seat belt is damaged or missing, report it to your supervisor immediately and do not operate the vehicle.

- In the event of an Industrial Vehicle tip over, seat belts shall not be released until the Industrial Vehicle motion has stopped. Operators should never attempt to jump free of an Industrial Vehicle that is tipping over.

5.13.34 Motorized Utility Industrial Vehicles (Cushman Carts, Golf Cart, Gators etc.)

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- Motorized utility Industrial Vehicles used for the transportation of employees, tools, parts and materials shall be equipped with seat belts for the operator and all passengers when supplied by the manufacturer.
- Operators and passengers shall keep all body parts inside the utility Industrial Vehicle while in motion.

5.14 Aerial and Scissor lifts

- 5.14.1 For Additional information and full scope of requirements reference CCS_HSS_STD_006.1 Aerial Lifts Safety Standard
- 5.14.2 Leased, rented or purchased aerial platforms shall be provided with Operator Controls (upper directional controls) at the platform, and minimally shall: Include an enabling device/control which:
- Automatically returns to the "off" or "neutral" position when released, if used to control any movement of the aerial platform. (Self-Centering type Controllers)
 - Be protected against activation other than that initiated by the operator.
 - Be clearly marked.
 - Operator protection in the form of a physical guard, pressure sensing device or cage to prevent the operator from being pinned between an overhead obstruction and the controls of the Boom-Supported Elevating Work Platform.
 - A presence sensing device (JLG - Skyguard, Genie – OPA, etc.) which will, at a minimum, sound the horn and stop all functions if the operator contacts the presence sensing device.
- 5.14.3 Leased, rented or purchased aerial platforms shall be equipped with clearly marked emergency lowering means readily accessible from ground level.
- 5.14.4 It is not permitted to operate the elevated platform with employees on the platform from the lower controls except to lower in an emergency situation.
- 5.14.5 Inspections
- 5.14.6 Pre-use inspection shall be performed prior to using the aerial platform prior to each operators use. The inspection shall be in accordance with items specified for owners and users of the aerial platform.
- 5.14.7 Annual Inspection shall be performed by a person qualified as a mechanic on the specific make and model of the aerial platform. The inspection shall be in accordance with items specified by the manufacturer.
- 5.14.8 The operating manual and maintenance manual shall be provided and shall be stored in the weather resistant storage compartment on the aerial platform.

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- 5.14.9 Prior to operating an aerial platform, the operator shall be properly trained by a qualified trainer and shall be familiar with the controls and operations of the specific aerial platform being used.
- 5.14.10 Conduct a work zone inspection to evaluate potential hazards associated with the area in which the platform is to be used
- 5.14.11 All employees in the aerial lift platform and those serving as ground personnel must be trained to operate the aerial platform.
- 5.14.12 Minimum training must include safe operator training on the type of aerial work platform / scissor lift.
 - (Including but not limited to an explanation of electrical, fall and falling object hazards; recognizing and avoiding unsafe conditions in the work environment, instructions for correct operation of the lift, demonstrations of the skills and knowledge needed to operate an aerial lift; when and how to perform inspections; any manufacturer requirements) as well as the proper selection and use of fall protection when using an aerial platform / scissor lift.

5.15 Use of Tape for Hazard Demarcation

- 5.15.1 Red danger tape shall be used to limit access to areas where an imminent hazard has been identified or where hazardous activity is taking place.
- 5.15.2 Only Authorized Employees may enter Red danger tape areas. For the purpose of this section the term “Authorized Employee” means:
 - Employees that have been assigned the task of eliminating the hazard.
- 5.15.3 All red tape must be tagged and identify the following:
 - Name and contact information of installer
 - Date of installation
 - Reason for installation
- 5.15.4 Yellow caution tape shall be used to identify hazards or activities that do not have imminent danger.
 - You may proceed with caution through yellow caution tape zones if you have a specific need to do so and are aware of the hazard.
- 5.15.5 All yellow tape must be tagged and identify the following:
 - Name and contact information of installer
 - Date of installation
 - Reason for installation

5.16 Walkways, Work Areas, Roadways, Stairs and Doors

- 5.16.1 Use stairways and walkways as provided. Do not take short cuts through stocking areas, over run-out tables or conveyors, or through any posted restricted areas. Go around or use cross-over walkways where they have been provided. Do not climb over handrails or equipment.
- 5.16.2 Designated storage/stocking areas shall be provided with adequate walkways to permit passage through and/or access to the materials in the storage/stocking areas.
- 5.16.3 When walking on or alongside roadways, to and from the gates, to your locker rooms or to your work location use designated walkways or approved routes.
- 5.16.4 Use handrails when going up or down stairs. If carrying an object, one hand must be free for the handrail. Walk, do not run. Do not go up or down more than one step at a time.
- 5.16.5 Do not descend stairs backwards unless the Job Safety Procedure for the job conditions requires it.
- 5.16.6 When walking along conveyor ramps and inclines maintain your balance and use handrails to assure that you do not come in contact with moving machinery or belts.
- 5.16.7 Hazardous locations shall be identified and precautions shall be taken to restrict access to unauthorized personnel.
- 5.16.8 All gates, doors or access points to the identified Hazardous Locations shall be posted with sign/signs that will communicate the following:
 - The specific hazard present.
 - Who is permitted to enter.
 - Procedures to be followed to enter.
 - Additional PPE/equipment required to enter.
- 5.16.9 Designated coil fields / coil staging areas shall be clearly delineated by high visibility paint, high visibility stanchions, and/or by chain, wire rope, rope, etc. In addition, permanent signs at entry points shall be posted.
 - Only properly trained and authorized personnel with the proper PPE are permitted to enter designated coil fields / coil staging areas.
- 5.16.10 Each plant shall develop and maintain Job Safety Procedures for entering and exiting a designated coil field, slab yard or staging area etc. which is serviced by a cab-operated crane, RC crane, and / or mobile equipment.

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5.16.11 Restricted Visibility Areas are locations where employees perform work and have restricted visibility from overhead cranes and / or mobile equipment. Examples are inside ladles, pits, under equipment, etc.

- Each department shall compile a list of designated Restricted Visibility Areas.
- Job Safety Procedures shall be developed, communicated for working within Restricted Visibility Areas which are serviced by a cab-operated crane, RC crane, and / or mobile equipment.

5.16.12 Inspection of Supported Walking and Working Surfaces.

- A documented monthly visual inspection including floor grating, floor plate, support structure, fasteners and welds etc.
- Immediately report any damage or defects associated with floor grating, floor plates, or associated support structures or securement/fasteners.
- Floor grating, floor plate, support structure, grating or plate fasteners and welds shall be given a physical inspection to be scheduled at a frequency not to exceed six months or immediately upon detection of damage, and/or substantial corrosion.
- Do not alter or modify floor grating, floor plate, support structure, grating or plate fasteners without authorization.

5.17 Excavations, Floors and Pit Openings

5.17.1 Unless authorized by your supervisor, do not start digging or excavating in any plant area and then, only after the completion of an excavation permit. Supervision must check with the plant Engineering Department for existence of pipes or conduits in the area to be excavated. Should the excavation come upon pipes, conduits, or red plastic tape, work must be stopped at once and supervision notified.

5.17.2 Excavation work 4 feet or deeper shall be considered a confined space and shall have a confined space permit or be dealt with appropriately.

- Every excavation which is four feet or deeper shall be continuously monitored with a gas monitor capable of monitoring at a minimum Oxygen, Carbon Monoxide, and L.E.L. (Lower Explosive Limit).

5.17.3 Employees making excavations, openings in floors or removing covers from sewers, manholes, valve pits, trenches, etc., are responsible for safeguarding them by erecting proper guards or barricades with signs.

- At night, warning lights shall be used to warn others of the danger.
- Deep trenches or excavations (over four feet) must be properly shored, sloped, or braced to guard against cave-in.

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- All excavated material must be piled at a minimum of three feet back from the edge of the excavation or designed by a qualified person.
- 5.17.4 Floor openings, such as pits, chutes, trap doors, manholes, platform edges, and hatchways, shall be guarded by a floor opening cover of standard strength and construction, or standard railing with standard toe boards.
- 5.17.5 Never step or jump over excavations, trenches, manholes, etc. Observe all barricades. Never remove a barricade unless authorized to do so.
- 5.17.6 Report any unguarded excavation, trench, open manhole, shaft, etc. to your supervisor immediately.

5.18 Working at Overhead Locations

- 5.18.1 If you are required to work overhead, you shall protect employees and/or contractors below by designating a hazardous work area on the ground or at any level below the work activity where there is potential for falling objects. Red Danger Tape (with identification, i.e., department name, contractor name) and Overhead Work Warning Signs shall be erected to completely encompass and block access to the Red Zone hazardous work area.
- 5.18.2 Entry into the Red Zone shall not be permitted until the following conditions are met:
- Entry into the Red Zone is absolutely required to complete the planned jobs/tasks.
 - A safety attendant must be in position to control entry into the Red Zone.
 - The attendant shall not permit entry into the Red Zone until the overhead work is stopped and communications are established between the employees on the ground/lower level and the crew working above.
 - Overhead work may continue after the communications have been established and the work procedures/protection are in place to protect the employees on the ground/lower levels.
- 5.18.3 A safety attendant is also required when entry into the Red Zone is not required but the hazardous work area cannot be completely encompassed with red danger zone tape.
- The overhead work area Red Zone entry permit must be completed, signed, and posted at the Red Zone entry along with all attached written procedures.
 - A safety attendant assigned to control entry into the Red Zone shall wear a fluorescent safety vest for identification and have an air horn or other device to warn unauthorized people away from the area.

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- Entry is permitted when the overhead work has stopped or completed and the "overhead" signs have been removed. If overhead work starts again, "overhead" signs and red danger tape shall be installed.

5.18.4 When jobs are planned, consideration must be given to overhead work and the potential for falling objects. Hole covers, nets, toe boards, screens, etc. must be considered and written in the preplan, JSA, and/or other work instruction.

5.18.5 The red danger tape shall be printed with Department or Contractor name, or signs may be attached to the tape to identify the Department or Contractor that established the Red Zone area.

5.18.6 Use approved ladders, scaffolding, or staging in good condition to reach overhead work.

- Report defects to your supervisor.
- Do not use improvised or makeshift arrangements.
- Loose materials, such as planks, scaffolding, pipe, bolts, nuts, tools, etc., must not be left overhead.
- Temporary scaffolding must be removed by authorized personnel at the conclusion of the job.
- Materials are not to be dropped or thrown to the ground but must be let down by hand lines or some other controlled method.

5.18.7 Overhead Work on Crane Runways

- Crane movements may present special hazards. The Crane Safety Standard, and other safe working procedures for the job being done must be followed while working on cranes or crane runways. This includes work on buildings, on top of tall bins, high machinery, and other areas where employees may be struck by the crane. See Crane Safety Standard for additional information.

5.18.8 Working on Roofs

- All work on roofs shall apply 100% fall protection for both edge protection and fall through protection. This includes traveling to the location on the roof where the work will be done. Exceptions to this rule will be:
- Roofs that have a certification by engineering that they are safe to walk on, and the work does not require edge protection.
- Office building roofs that are designed with edge protection.
- Roof access points, such as ladders or stairways, shall display signs stating "Access to Roof is prohibited without Department Authorization".

5.18.9 Working on Cranes

- Prior to commencing any preventative, breakdown, or shutdown maintenance on an overhead crane, a Red Pendant Flag (highly visible) must be lowered from the crane 10 to 15 feet above the floor. The pendant shall have "WARNING" displayed on both sides and shall easily be seen from the floor.
- Warning signs shall also be placed on the floor (i.e., "Caution Overhead Work") to alert others in the area of the work being performed on the crane overhead.
- The warning signs will be the maximum protection required if there is no potential for tools, parts, or materials to fall to the area below. Employees and contractors shall use caution when walking under any part of an overhead crane that has the red pendant "WARNING" flag displayed and warning signs on the floor indicating the crane is down for maintenance.
- A Red Danger Tape Zone (with identification, i.e., department name, contractor name) shall be erected to completely encompass and block access to work areas on the ground when there is a potential for tools, parts or materials to fall from the crane during the maintenance activity. Entry into the Red Zone shall be prohibited during the maintenance activity.

5.19 Ladders, Scaffolding and Staging

5.19.1 Extension Ladders

- Always inspect ladders before use. Defective ladders shall be tagged and immediately removed from service. Notify your supervisor for a replacement.
- Extension ladders are required to be inspected annually and documented.
- Extension ladders must be labeled with load capacity clearly marked and visible
- Portable metal ladders including aluminum shall not be approved for use.
- Never paint or use painted wooden ladders. Paint will hide cracks, knots, rot, or other defects.
- Straight portable ladders and extension ladders shall extend at least three feet above the top access level and be placed so the distance from the base of the ladder to the wall or other supporting structure is about one-quarter the length of the ladder from the floor to the top support point. These ladders are to be tied in place.
- Both rails of the extension ladder must be tied off to a structure capable of stabilizing and supporting the ladder from falling. If the ladder cannot be tied off it must be held by a coworker whenever someone is working on the ladder.

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5.19.2 When working from straight, extension, or non-self-supporting portable ladders and over 4 feet from the ground you must wear a full body harness fall arrest system and be tied-off to a proper anchor point unless the following conditions are met:

- The worker can maintain a 3-point touch, 2 hands and one foot, two feet and one hand, or two feet and the midline of the body. Examples are:
- Both feet and 1 hand - on the ladder, belt buckle or midline of the body must be between the rails of the ladder, and the shoulders below the top rail. Any stretching for position requires repositioning of the ladder.
- Both legs and body leading against the ladder. The belt buckle/midline of the body must be between rails. Any stretching for position perpendicular to the ladder requires repositioning of the ladder. Feet must be firmly on the ladder and the belt buckle/midline of the body must be resting on the ladder. The shoulders must be below the top rung. Worker may use two hands for short duration work (changing light bulbs, opening hand valves, etc.).

5.19.3 Step ladders

- Step ladders are required to be inspected annually and documented.
- Step ladder must be labeled with load capacity clearly marked and visible
- Shall be fully opened and locked.
- Never stand on top of a step ladder.
- Work from a point at least two feet or two steps from the top.

5.19.4 When climbing any ladder (portable or stationary),

- Hold on with both hands and always face the ladder.
- Use a hand line to raise or lower tools that you will need.
- Never exceed the maximum intended load of the ladder.

5.19.5 Scaffolding & Staging.

5.19.6 Employees who perform work while on a scaffold shall be trained by a qualified person to recognize the hazards associated with the type of scaffold being used, the nature of any electrical hazards, fall hazards, and falling object hazards in the work area.

5.19.7 Any scaffold damaged or weakened from any cause shall be immediately repaired or replaced and shall not be used until repairs have been completed.

5.19.8 Scaffolds shall not be loaded in excess of the working load for which they are intended.

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- 5.19.9 Tools, materials, and debris shall not be allowed to accumulate in quantities to cause a hazard when working on a scaffold.
- 5.19.10 All scaffold planking must be approved and identified as scaffold grade.
- 5.19.11 Access to a scaffold shall be by secured ladder, or an equivalent safe access shall be provided. Climbing the side of scaffolding is not permitted.
- 5.19.12 Scaffolding greater than four feet in height must be equipped with handrails, mid-rails, toe boards, and be tightly planked.
- 5.19.13 Fall protection equipment including a full fall arrest system and retractable lanyard shall be worn and connected to an approved anchorage point or lifeline when working on any scaffold platform greater than four feet in height and not equipped with standard top rails and mid-rails or is not tagged
- 5.19.14 A scaffold tag signed by a competent person is required prior to allowing personnel to work on scaffolds. The tag certifies proper construction and that safety features are in place. The scaffold inspection checklist must be completed before accessing the scaffold.

5.20 Safe Storing, Loading and Manual Handling of Material

- 5.20.1 Follow the safe practice in your department for blocking materials to prevent rolling or falling. Check blocking to make sure that removing material will not result in a spill or shifting in other parts of the pile.
- 5.20.2 Avoid overloading shelves and overhead storage areas. Never store materials on shelves, roofs, or cabinets unless they are rated, and load posted.
 - In operating areas shelves, cabinets, lockers, toolboxes, etc. must be secured to prevent tip-over.
- 5.20.3 Storage capacity must be marked on all shelving designed for industrial storage.
- 5.20.4 Items in storage that weigh 50 pounds or more should be secured to a pallet and accessible to a forklift or other mechanical means for safe handling and storage.
- 5.20.5 Employees are not permitted to climb on piled sheets, slabs, scrap, coal, coke, plates, coils, pipe, or other material unless it is required by their job duties and documented in a Job Safety Procedure.
- 5.20.6 Loading & Unloading
- 5.20.7 Know the safe load limits and the approved methods of securing loads of piled materials. Never load any material handling equipment such as trailers, trucks, buggies, elevators, etc., beyond its safe limit. Tie or band material if necessary to prevent shifting.

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- 5.20.8 Always check the condition of the floor of trucks and railroad cars for holes or weak spots. Report defective cars or trucks to your supervisor before working in them.
- 5.20.9 Truck beds or railroad car beds that contain debris (i.e.: paper, plastic, dunnage, etc.) must be cleaned to allow inspection of the integrity of the floor before beginning work.
- 5.20.10 Employees shall not walk or work on a truck bed or in a railcar while stepping on paper, plastic, or any other debris.
- 5.20.11 Only properly trained and authorized personnel are permitted to load or unload tank cars and trucks containing flammable liquids. Open flames, spark producing devices and unprotected electrical devices must be removed from the area before proceeding with the loading or unloading of flammable liquids. Specific job procedures must be developed for loading and unloading of each chemical.
- 5.20.12 Stay away from all tank cars or trucks if you are not trained and equipped to work with the potentially hazardous chemical that they might contain.
- 5.20.13 Manual Material Handling
- No employee will lift more than 50 pounds.

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7 Revision History

Rev.	Date	Section	Paragraph	Summary of change	Authorized by
1	11-14-2022	Entire document supersedes all previous	Entire document supersedes all previous	Initial issue	K. D. Berard